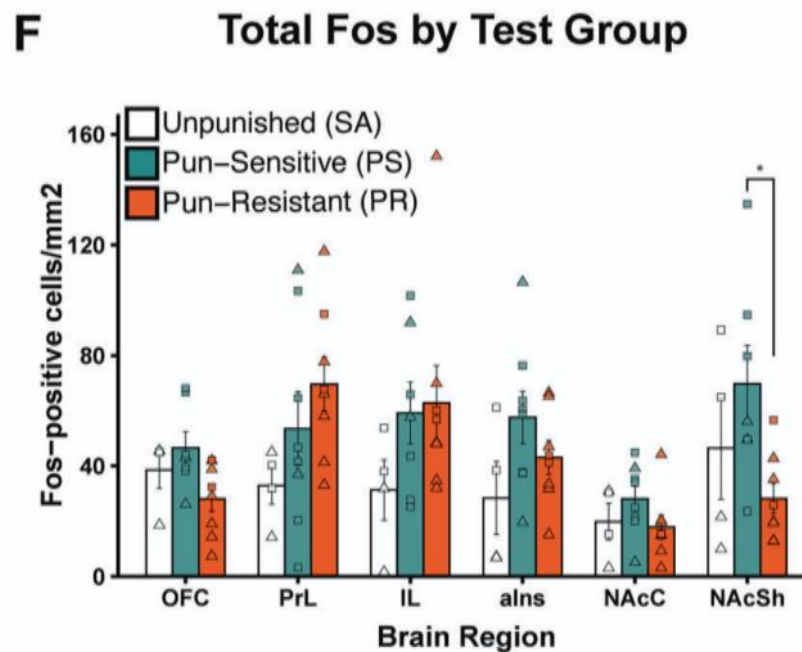


Article Critique #3

Alcohol use disorder (AUD) is defined by the loss of control over alcohol consumption, despite any/all negative consequences. While AUD only affects a small portion of the population, it is a pervasive and destructive disorder that we are still all too unfamiliar with. More specifically, we are totally ignorant as to how this disorder manifests itself. Fortunately, the results of previous studies have been successful in identifying some AUD-correlative regions (in both human and mice): clinical studies often modulate cortico-striatal circuits, human DBS studies support the VS as a promising target area, and mice studies also targeting cortico-striatal circuits (specifically: medial prefrontal cortex to Nucleus Accumbens, anterior Insular Cortex to Nucleus Accumbens, and dopamine input to anterior dorsolateral striatum). However, each one of these preliminary studies was performed with exclusively male rats, which is obviously an oversight that needed to be addressed, especially considering the fact that AUD rates are increasing faster in women than they are in men. For this reason, the researchers intentionally included female rats in their study to identify the specific role of the nucleus accumbens shell (NAcSh) in mediating this “punishment-resistance” addiction behavior.

To see exactly how punishment-resistance (a hallmark of alcohol use disorder) is represented in the brain, the researchers utilized 96 outbred Long-Evans rats (48 male, 48 female). The experiment began with a four-week intermittent alcohol access protocol in the home cage, where rats had 24-hour access to 20% ethanol three times per week to establish a high baseline intake. Following this, the rats were moved to operant chambers for training. They progressed from simple fixed-ratio 1 (FR-1) and FR-2 schedules to a FR-3 schedule, meaning they had to press a specific lever three times to receive a 0.12 mL dose of 20% alcohol. To model the "negative consequences" seen in human addiction, the team introduced a response-contingent punishment phase. During these FR-3 sessions, there was a 33% probability that the second lever press in the sequence would trigger a foot-shock. The intensity of the shock was gradually increased from 0.20 mA to 0.30 mA. To quantify how much each rat suppressed its drinking to avoid pain, the researchers calculated a Suppression Ratio (SR): $\text{Punished Responding} / (\text{Baseline} + \text{Punished Responding})$. A rat with an SR of 0.5 showed no suppression (punishment-resistant), while a rat with a score near 0 stopped almost entirely (punishment-sensitive). The results revealed a significant sex difference: female rats performed more lever presses, consumed more alcohol, and were far more likely to be categorized as punishment-resistant than males. To find the neural "signature" of this resistance, the team used cFos as an indirect marker of neuronal activity. As shown in Figure 2F, they analyzed multiple regions, including the orbitofrontal cortex and anterior insula, but the NAcSh was the only region that showed a significant difference. Specifically, punishment-resistant rats exhibited significantly lower cFos expression in the NAcSh compared to sensitive rats during the punished test. Finally, to prove that this NAcSh hypofunction actually caused the behavior, the researchers used chemogenetics.

They bilaterally injected an inhibitory DREADD (hM4Di) into the NAcSh of a new group of rats. When they administered the ligand deschloroclozapine (DCZ) to shut down the NAcSh, they found it had no effect on unpunished drinking or general motivation (as measured by a progressive ratio test). However, in the 0.30 mA punishment sessions, inhibiting the NAcSh selectively increased alcohol intake specifically in the punishment-resistant female rats. This confirmed that NAcSh activity is a critical "brake" on compulsive intake in females.



As with all research, there are a few strengths and flaws here that are worth addressing (in the interest of future research endeavors and treatment developments for AUD). First, using chemogenetics was a major strength as it provided a level of causal evidence that simple cFos imaging just cannot reach on its own. Second, I appreciate the fact that they preserved the Punishment-Resistance Score as a continuous variable for their main stats rather than just relying on arbitrary "cut-off" points, which makes the data much more reliable. Third, finally including females in a punishment model is a major win for the field, and the specific results they found support the theory that females are more susceptible to AUD. However, there are some limitations here too that cannot be ignored. The researchers admitted they were statistically underpowered to fully compare sex differences in cFos activity because they simply didn't have enough punishment-resistant males to form a viable comparison group. Additionally, the lack of a yoked-shock control means we cannot be 100% certain if the low NAcSh activity was a direct result of the psychological "choice" to drink or just a reaction to the physical sensation of the shock. Finally, the NAcSh is a highly heterogeneous region with different functions along its

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anatomical axes, so a more localized sub-region analysis would be necessary to fully map these effects.

Although many people might not be focused on specific lab data regarding rodent alcohol consumption, the implications of this study are massive for everyone, particularly clinical psychiatrists and those working in women's health. Since the gender gap in AUD is closing so rapidly, these results offer two vital insights for the medical community. First, by identifying NAcSh hypofunction as a primary driver for aversion-resistant drinking, we now have a localized neural target for future treatments like deep brain stimulation. Second, the discovery that punishment-resistance and motivation are orthogonal suggests that we need to move away from treating addiction as a single, uniform condition. This research takes us one step closer to developing sex-specific interventions ("personalized medicine") that actually address the unique ways that women experience compulsion and relapse.